

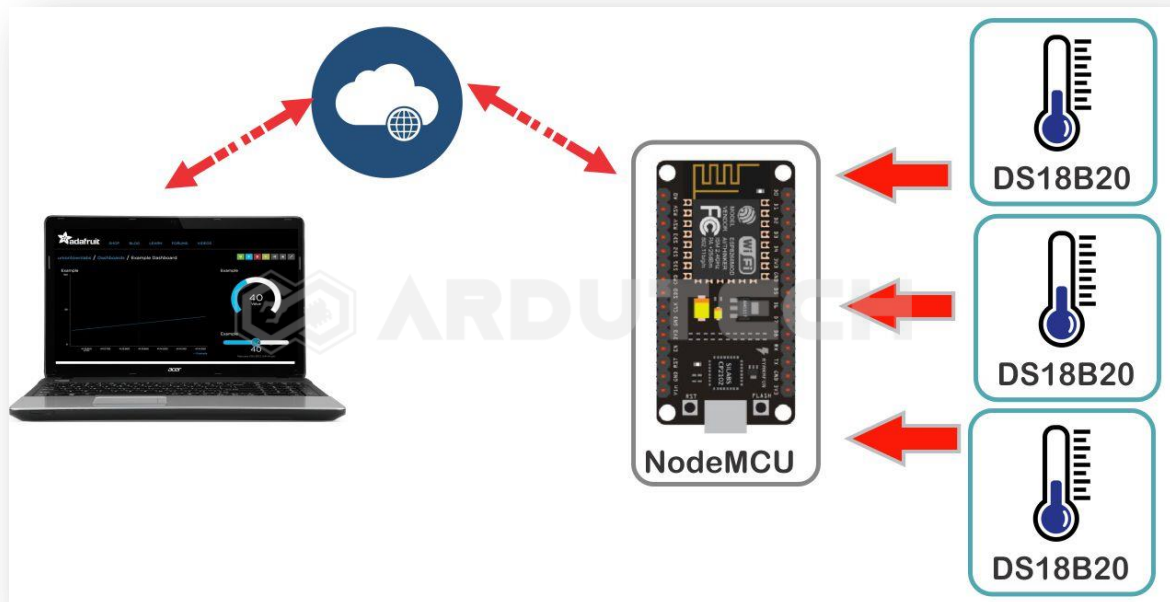
3 Channel DS18B20 Adafruit.io

Deskripsi

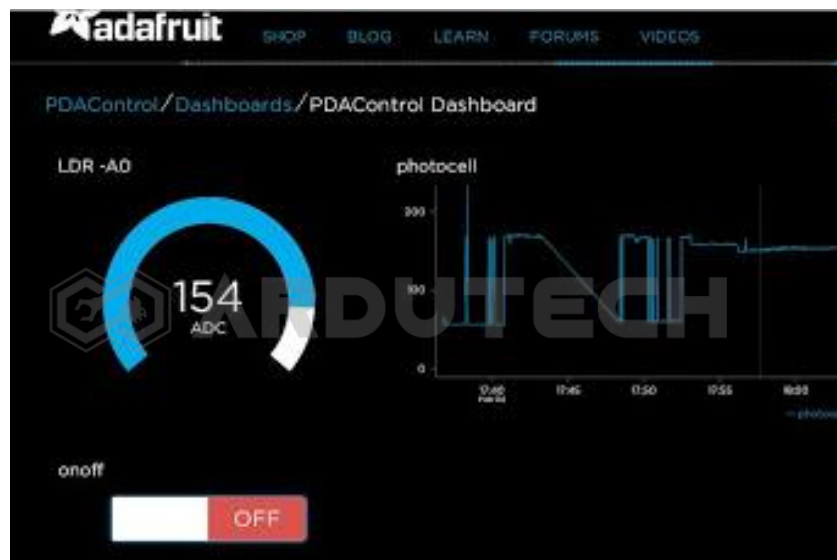
Membuat proyek IoT (*Internet of Things*) untuk membaca data 3 sensor suhu DS18B20 kemudian data dikirim secara kontinyu ke Adafruit IO dengan protokol MQTT.

Cara Kerja

Sensor suhu dengan output digital yaitu DS18B20 membaca suhu lingkungan kemudian data dikirim oleh NodeMCU melalui jaringan internet ke Adafruit secara kontinyu, pada contoh proyek ini setiap 5 detik, dapat diedit pada programnya.



Adafruit IO adalah sistem yang disediakan oleh Adafruit (pembuat modul – modul sensor, perangkat elektronik, robot dll) untuk membuat sebuah proyek/aplikasi IoT (*Internet of Things*). Adafruit IO ini boleh dipakai oleh siapa saja karena sifatnya free.



Sensor suhu DS1820B merupakan sensor dengan output digital, berbeda dengan sensor suhu LM35 yang keluarannya analog. Karena output-nya digital sehingga cara membaca hasil *sensing*-nya juga secara digital yaitu dengan komunikasi yang memakai protokol *one – wire*. Cukup 1 jalur data saja yang terhubung dengan mikrokontroler (NodeMCU).

Sensor ini mampu mengukur suhu dengan range -55 sampai dengan 125⁰C. Di dalamnya terdapat ADC (Analog to Digital Converter) dengan resolusi 12 bit sehingga mempunyai kemampuan membaca perubahan temperatur s/d 0,5⁰C.



Sensor DS18B20 terdiri dari 3 pin/kaki sehingga menyederhanakan dalam rangkaiannya. Cukup sambungkan dengan tegangan untuk 2 kaki (Vcc dan Gnd) dan satu pin untuk jalur data.

Spesifikasi sensor suhu DS18B20 :

- Interfacing dengan protokol 1-Wire
- Addressable dg alamat/kode 64 bit
- Tegangan input 3.0V hingga 5.5V
- Range pengukuran dari -55°C hingga +125 °C
- Akurasi +/-0.5 °C pada rentang -10 °C hingga +85 °C
- Resolusi sensor bisa dipilih mulai dari 9 hingga 12 bit

Kebutuhan Bahan

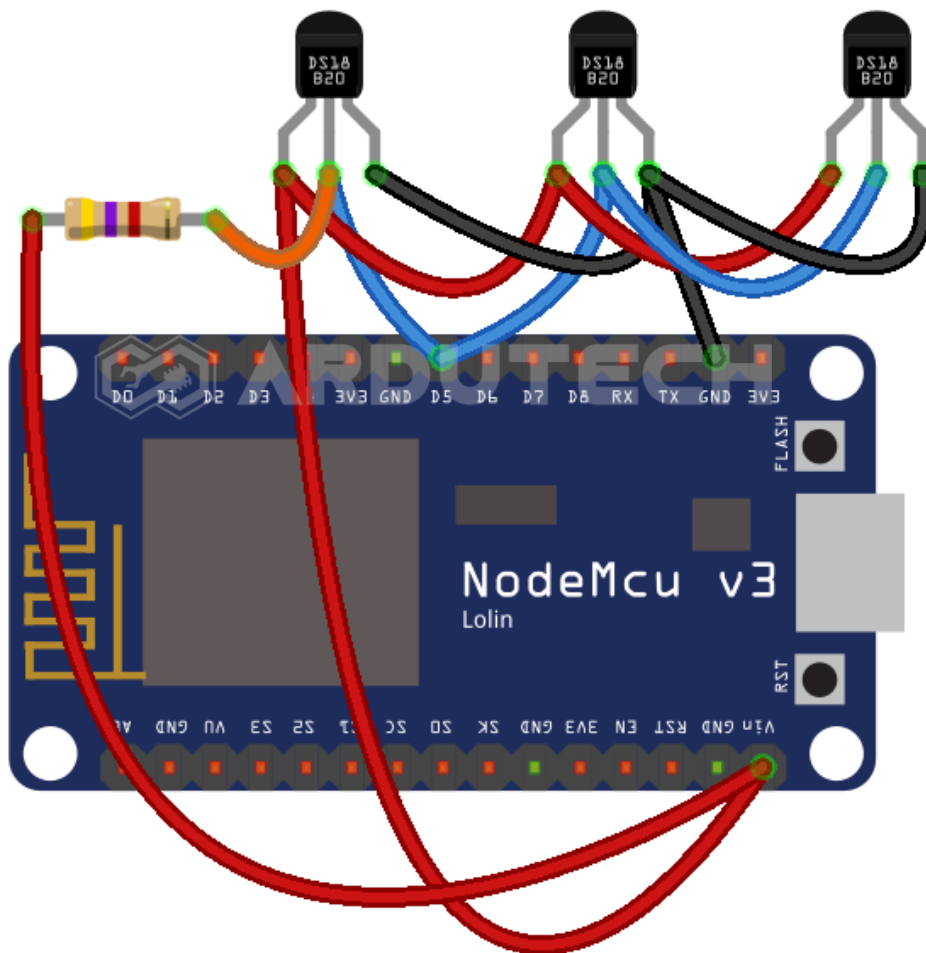
- NodeMCU V3
- Sensor DS18B20 (3 buah)
- Resistor 4K7
- Kabel konektor
- Kabel micro USB



Kebutuhan Software

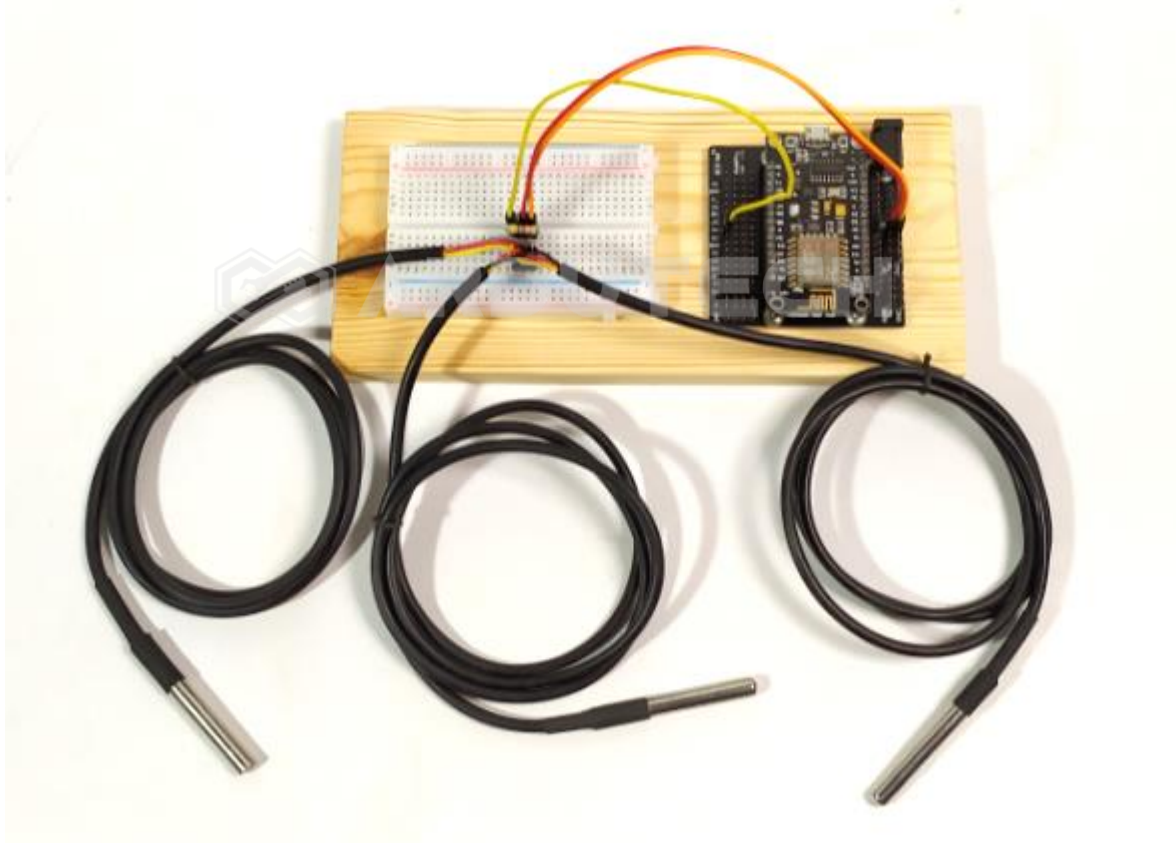
- Arduino IDE
- Adafruit IO

Rangkaian/Skematik



Koneksi NodeMCU dengan Sensor DS18B20 :

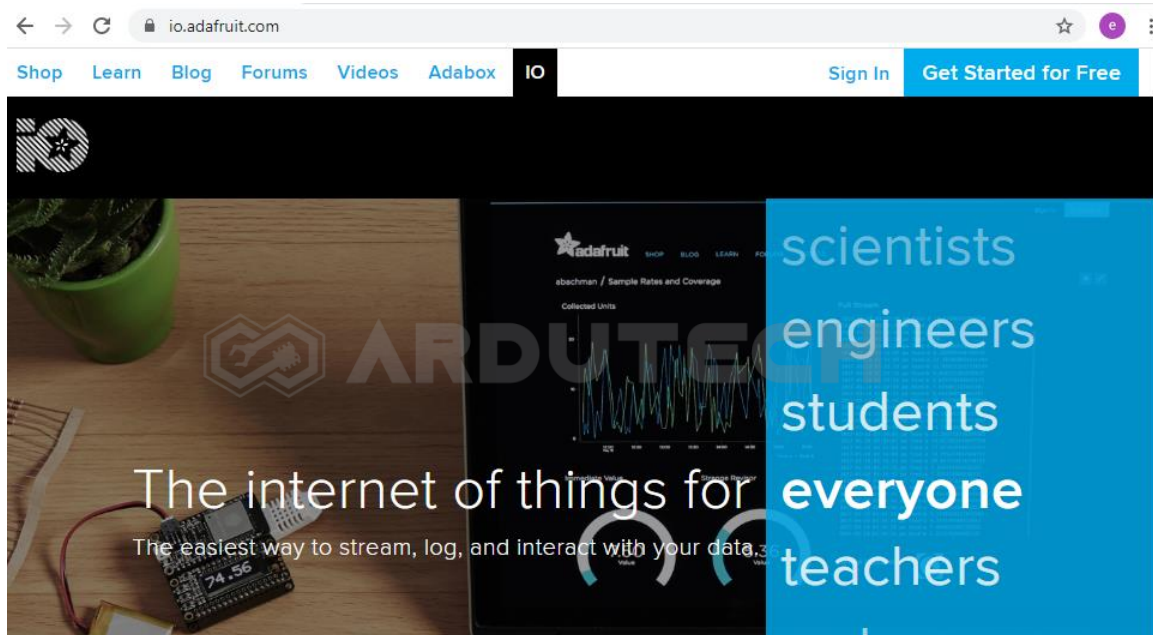
NodeMCU	DS18B20
D5	OUT
Vin (5V)	VCC
GND	GND



Tambahkan resistor pull-up 4K7 ohm antara pin Data dengan VCC.

Membuat Antarmuka di Adafruit

Silakan masuk ke halaman <https://io.adafruit.com/>



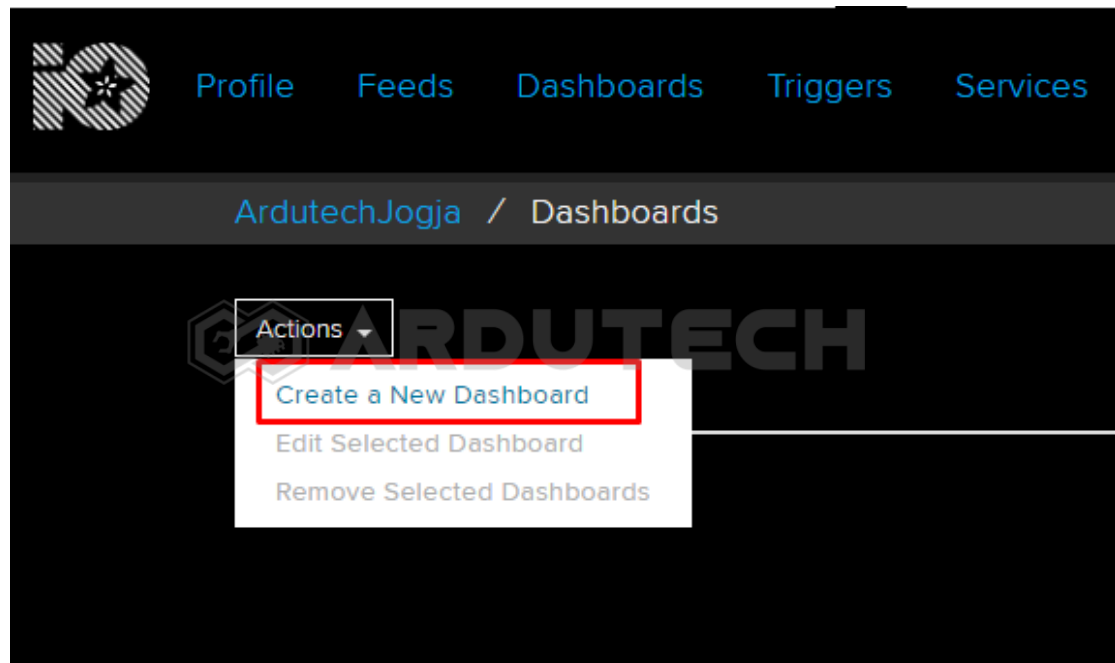
Jika belum mempunyai akun silakan membuat akun terlebih dahulu. Siapkan sebuah akun email aktif.

Klik pada **“Get Started for free”** di bagian pojok kanan atas.

Isi data – data yang diperlukan kemudian klik **“CREATE ACCOUNT”** dan selanjutnya ikuti petunjuknya.

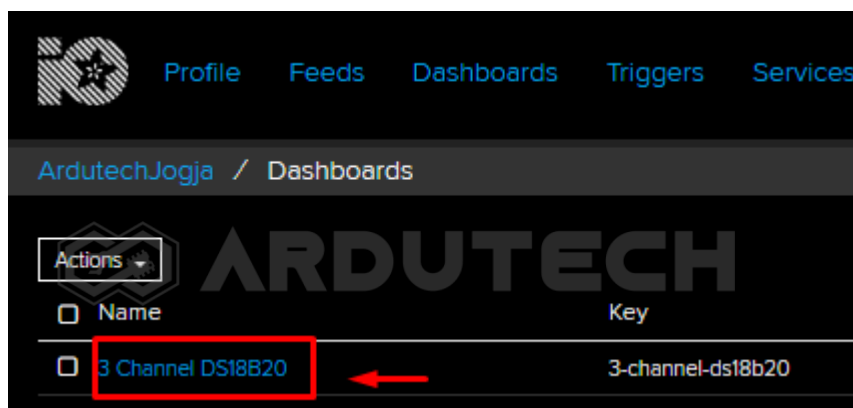
Masuk ke menu **IO**.

Klik pada tombol **“Actions”** kemudian pilih **“Create a New Dashboard”**

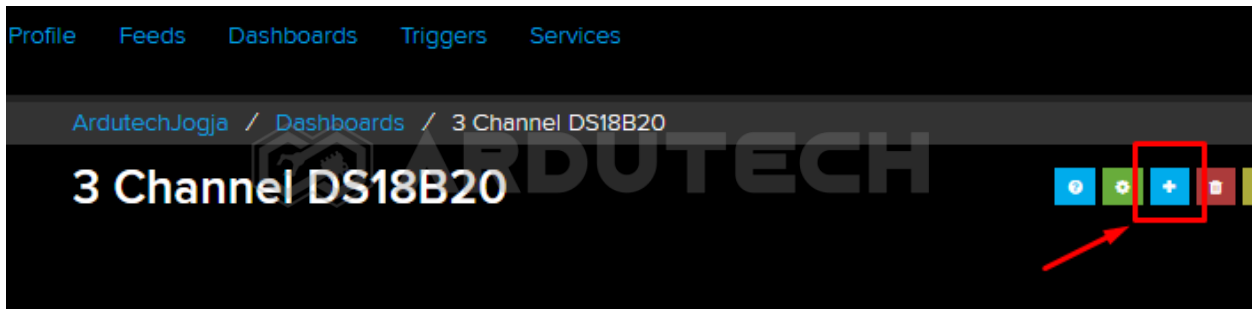


Selanjutnya akan muncul jendela untuk memberi nama, silakan beri nama pada dashboard yang akan dibuat, misalnya **“3 Channel DS18B20”** dan klik **“Create”**.

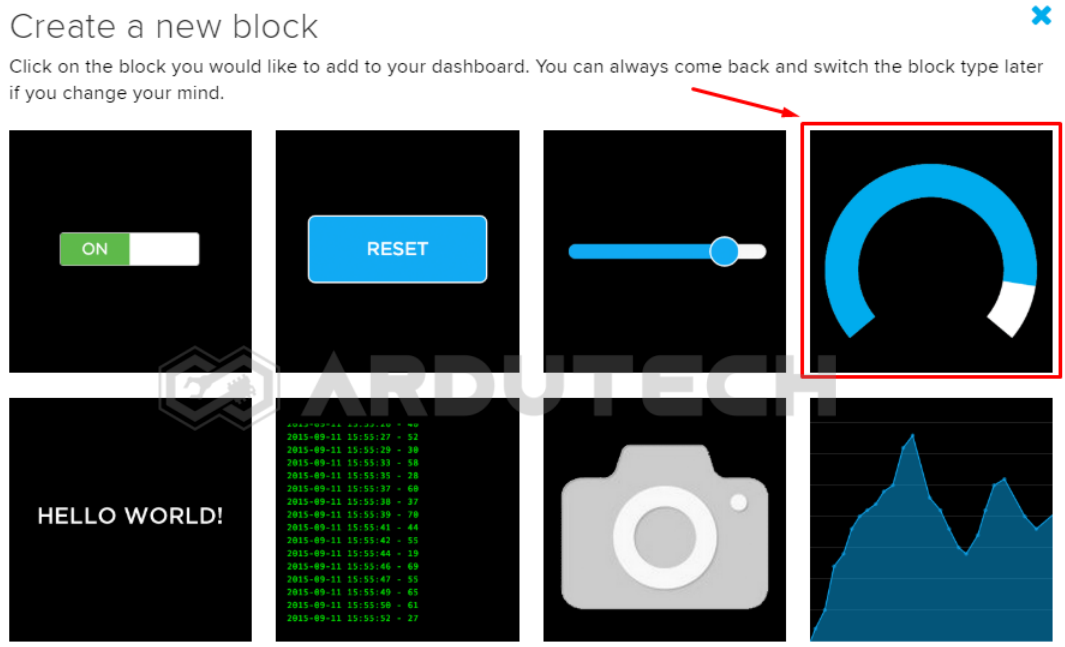
Setelah klik tombol **“Create”**, akan muncul dashboard baru dengan nama **“3 Channel DS18B20”**.



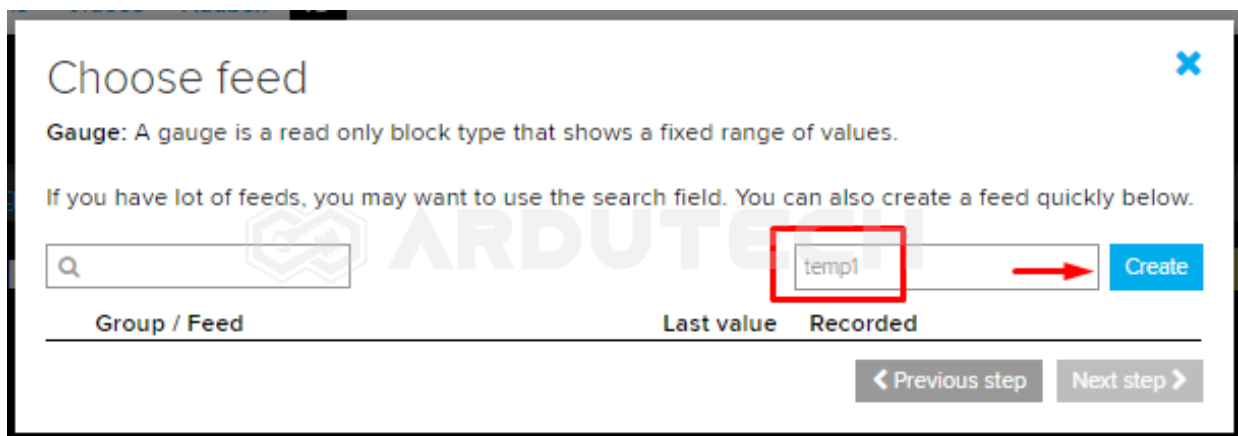
Klik dashboard **“3 Channel DS18B20”** kemudian pilih tombol **“+”** untuk membuat **block** baru.



Setelah tombol “+” di-klik akan muncul tampilan bermacam ‘widget’. Ada tombol (Toggle), tampilan Gauge, Slider, Text dll. Pilih **Gauge** (pojok kanan atas).



Selanjutnya muncul tampilan untuk memilih ‘Chose feed’ . Ini semacam “Topik” dalam komunikasi MQTT. Jadi supaya dikenali oleh ‘subscriber’ (penerima perintah) maka ‘publisher’ harus menentukan ‘topik’ nya. Silakan tulis “temp1” kemudian klik tombol “Create”.



Muncul feed **temp1**.

Choose feed

Gauge: A gauge is a read only block type that shows a fixed range of values.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Group / Feed	Last value	Recorded
☒ temp1	🔒	1 minute

Pilih Feed **temp1** selanjutnya klik **"Next Step > "** sehingga muncul tampilan **Block settings** :

Block settings

In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the "Create Block" button to send it to your dashboard.

Block Title (optional)

Gauge Min Value

Gauge Max Value

Gauge Width

Gauge Label

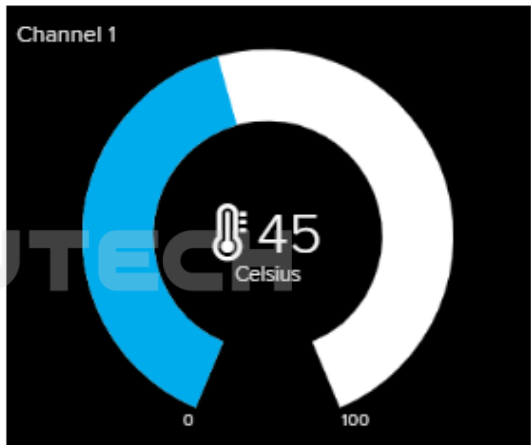
Low Warning Value

Optional. If no low warning value is given, the gauge will only change color when the value is out of bounds.

High Warning Value

Block Preview

Channel 1



Gauge A gauge is a read only block type that shows a fixed range of values.

Test Value

Decimal Places

2

Number of decimal places to display when value is a number. Defaults to 2.

☒ Show Icon

When checked, show an icon with the value.

Icon

thermometer

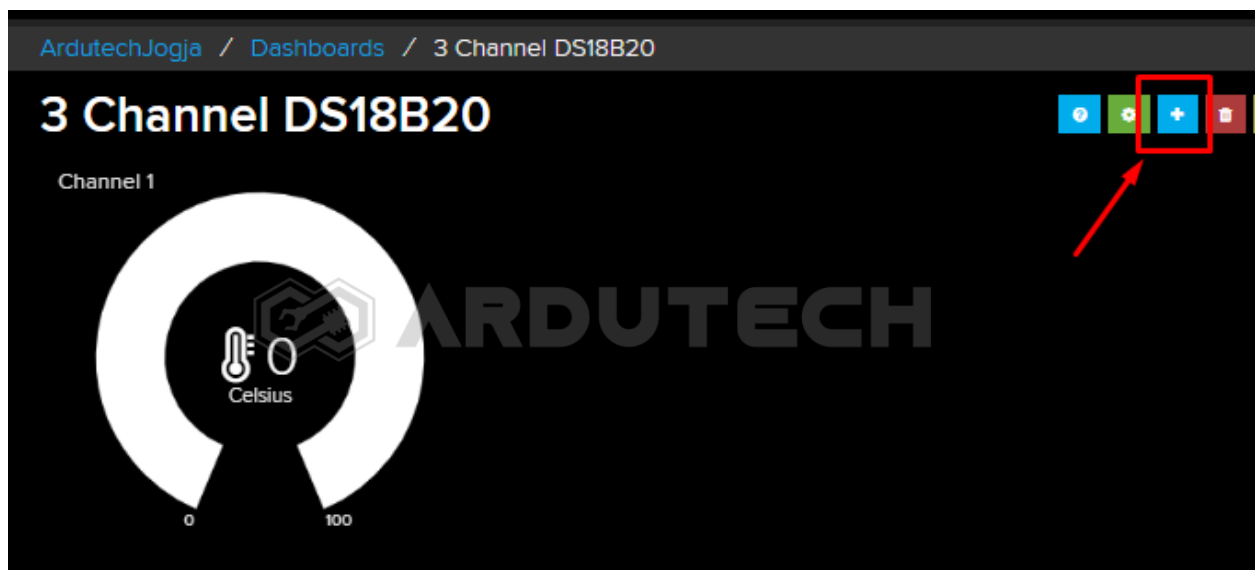
Show this icon next to the value.

< Previous step Create block

Isi pada kolom yang tersedia :

- Block Title : Channel 1 (opsional)
- Gauge Min Value : 0
- Gauge Max Value : 100
- Gauge Width : 50px
- Gauge Label : Celsius

Klik tombol **“Create block”**. Selesai pembuatan Block Channel 1 (Sensor 1 DS18B20) di Adafruit.IO. Kita masih perlu 2 Block lagi untuk 2 sensor yang lain.



Klik tombol plus “+” lagi untuk menambahkan sebuah Gauge lagi ke dashboard.

Setelah tombol “+” di-klik akan muncul tampilan bermacam ‘widget’. Ada tombol (Toggle), tampilan Gauge, Slider, Text dll. Pilih **Gauge** (pojok kanan atas).

Create a new block

Click on the block you would like to add to your dashboard. You can always come back and switch the block type later if you change your mind.



Selanjutnya muncul tampilan untuk memilih 'Chose feed'. Silakan tulis **"temp2"** kemudian klik tombol **"Create"**.

Choose feed

Gauge: A gauge is a read only block type that shows a fixed range of values.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Search:

Group / Feed	Last value	Recorded
☐ temp1	5 minutes	

Muncul feed temp2. Klik /pilih **temp2**.

Choose feed

Gauge: A gauge is a read only block type that shows a fixed range of values.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Search:

Group / Feed	Last value	Recorded
☐ temp1	5 minutes	
☒ temp2	1 minute	

Selanjutnya klik **"Next Step > "** sehingga muncul tampilan **Block settings** :

Block settings

In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the "Create Block" button to send it to your dashboard.

Block Title (optional)

Gauge Min Value

Gauge Max Value

Gauge Width

Gauge Label

Low Warning Value

Optional. If no low warning value is given, the gauge will only change color when the value is out of bounds.

High Warning Value

Optional. If no high warning value is given, the gauge will only change color when the value is out of bounds.

Decimal Places

Number of decimal places to display when value is a number. Defaults to 2.

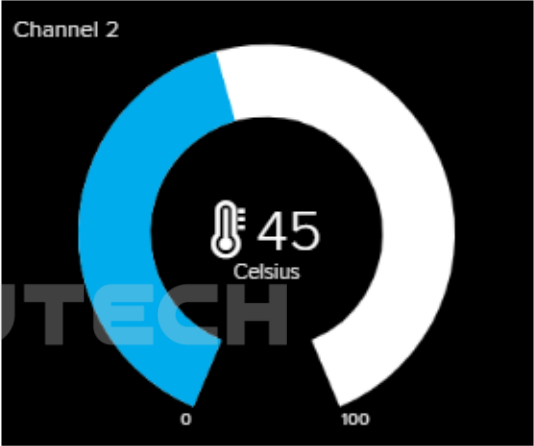
☒ Show Icon

When checked, show an icon with the value.

Icon

Show this icon next to the value.

Block Preview



Gauge A gauge is a read only block type that shows a fixed range of values.

Test Value

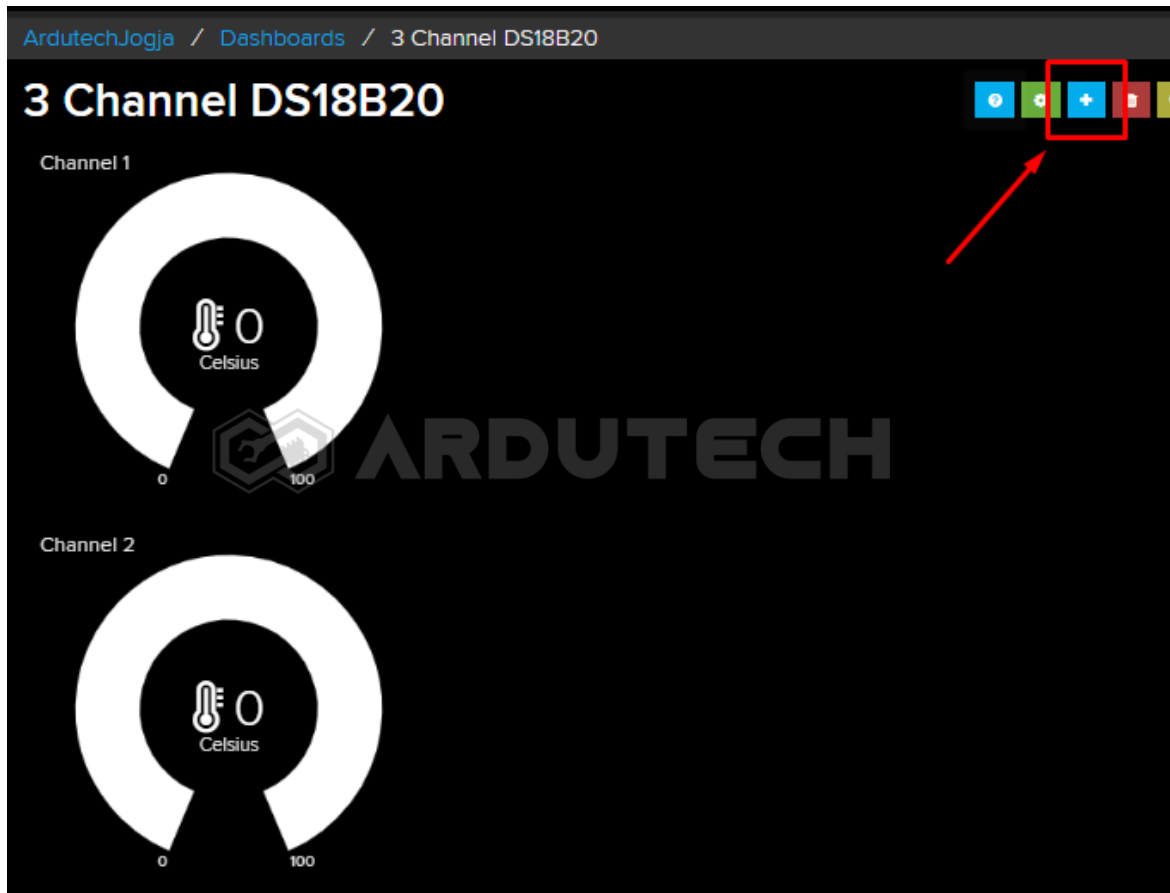
< Previous step

Create block

Isi pada kolom yang tersedia :

- Block Title : Channel 2 (opsional)
- Gauge Min Value : 0
- Gauge Max Value : 100
- Gauge Width : 50px
- Gauge Label : Celsius

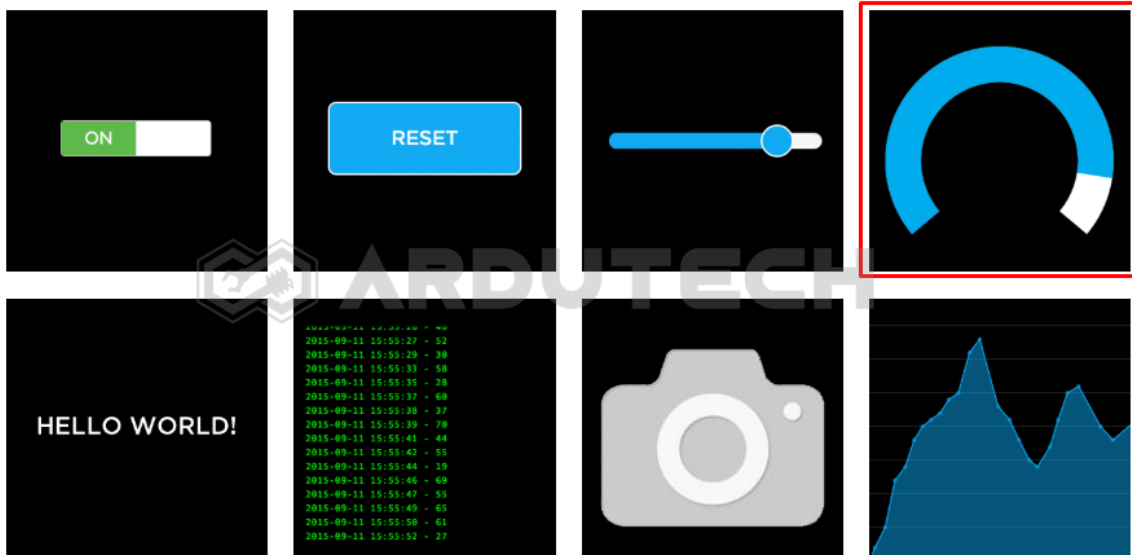
Klik tombol “**Create block**”. Selesai pembuatan 2 Block di Adafruit.IO. Masih perlu 1 Block lagi untuk sensor ke-3.



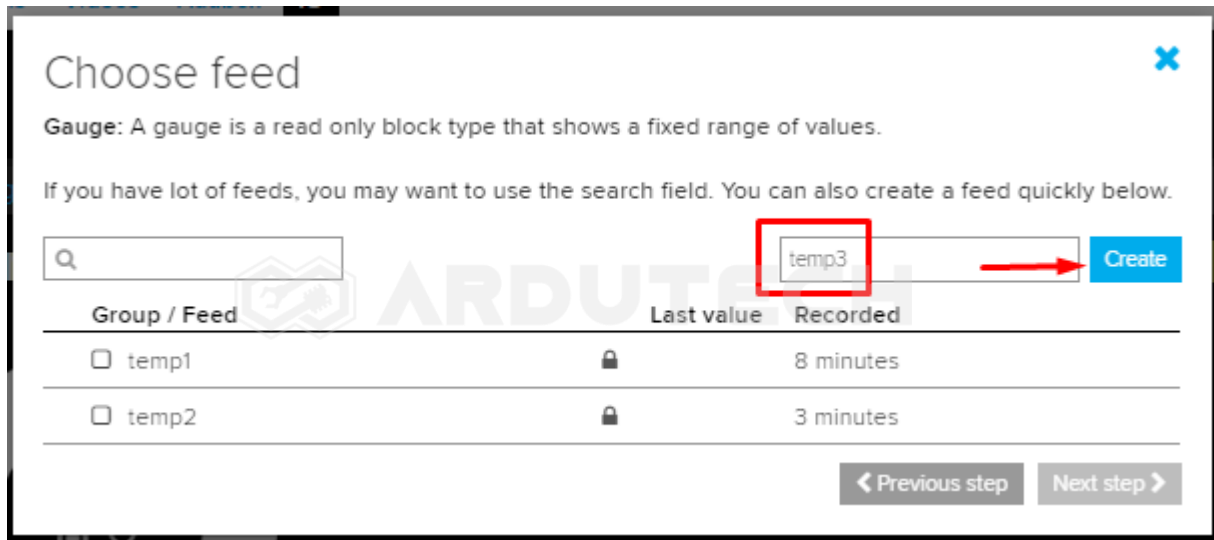
Klik tombol plus “+” lagi untuk menambahkan sebuah Gauge lagi ke dashboard. Setelah tombol “+” di-klik akan muncul tampilan bermacam ‘widget’. Ada tombol (Toggle), tampilan Gauge, Slider, Text dll. Pilih **Gauge** (pojok kanan atas).

Create a new block

Click on the block you would like to add to your dashboard. You can always come back and switch the block type later if you change your mind.



Selanjutnya muncul tampilan untuk memilih 'Chose feed' .Silakan tulis “temp3” kemudian klik tombol “Create”.



Choose feed

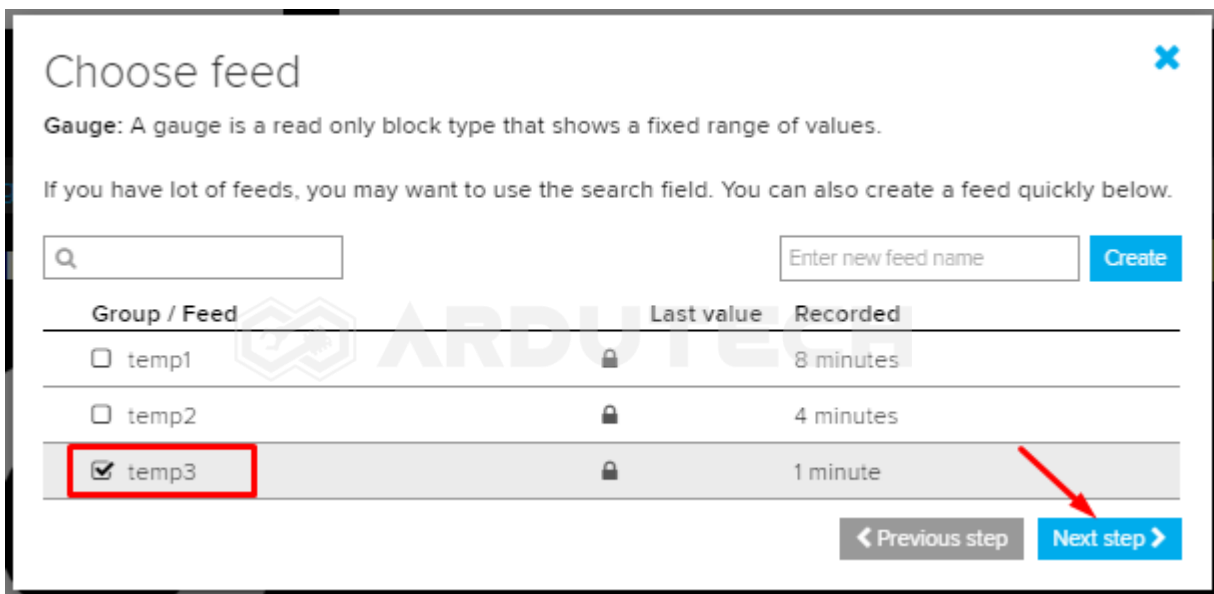
Gauge: A gauge is a read only block type that shows a fixed range of values.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Search:

Group / Feed	Last value	Recorded
☐ temp1	🔒	8 minutes
☐ temp2	🔒	3 minutes

Muncul feed temp3. Klik /pilih **temp3**.



Choose feed

Gauge: A gauge is a read only block type that shows a fixed range of values.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Search:

Group / Feed	Last value	Recorded
☐ temp1	🔒	8 minutes
☐ temp2	🔒	4 minutes
☒ temp3	🔒	1 minute

Selanjutnya klik “Next Step >” sehingga muncul tampilan **Block settings** :

Block settings

In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the "Create Block" button to send it to your dashboard.

Block Title (optional)

Gauge Min Value

Gauge Max Value

Gauge Width

Gauge Label

Low Warning Value

Optional. If no low warning value is given, the gauge will only change color when the value is out of bounds.

High Warning Value

Optional. If no high warning value is given, the gauge will only change color when the value is out of bounds.

Decimal Places

Number of decimal places to display when value is a number. Defaults to 2.

☒ Show Icon

When checked, show an icon with the value.

Icon

Show this icon next to the value.

Block Preview

Gauge A gauge is a read only block type that shows a fixed range of values.

Test Value

< Previous step

Create block

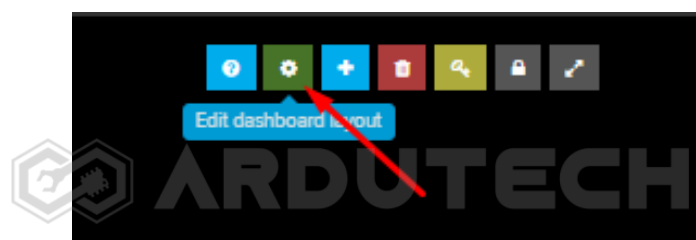
Isi pada kolom yang tersedia :

- Block Title : Channel 3 (opsional)
- Gauge Min Value : 0
- Gauge Max Value : 100
- Gauge Width : 50px
- Gauge Label : Celsius

Klik tombol “**Create block**”. Selesai pembuatan antarmuka di Adafruit.IO.



Jika ingin mengedit penempatan (layout) tinggal klik tombol **edit**.



Kemudian atur tata letaknya, setelah selesai klik tombol **Save**.



Selanjutnya kita perlu kode (token) untuk mengisi variabel username dan active key di program Arduino yang akan dibuat. Klik bagian “**Adafruit IO Key**” di pojok kanan atas.

YOUR ADAFRUIT IO KEY

Your Adafruit IO Key should be kept in a safe place and treated with the same care as your Adafruit username and password. People who have access to your Adafruit IO Key can view all of your data, create new feeds for your account, and manipulate your active feeds.

If you need to regenerate a new Adafruit IO Key, all of your existing programs and scripts will need to be manually changed to the new key.

Username:

Active Key:

[Hide Code Samples](#)

Arduino

```
#define IO_USERNAME "ArdutechJogja"
#define IO_KEY "ruu_wUXJ670DJNEIIBQWERTY567MNB"
```

[REGENERATE KEY](#)

Perhatikan untuk **Username** dan **Active Key** ini penting, karena nanti dipakai ketika pemrograman dengan Arduino IDE. Catat/simpan kode tersebut.

Nah untuk pembuatan antarmuka pada Adafruit sudah selesai, sekarang kita siapkan program Arduino IDE-nya.

Program/Source Code

Program pada proyek ini memerlukan library :

- ESP8266WiFi.h
- Adafruit_MQTT.h
- Adafruit_MQTT_Client.h
- OneWire.h
- DallasTemperature.h

Buka/jalankan Arduino IDE kemudian buat lembar kerja baru. Tulis kode program berikut.

```

/*****

* Project 3 Channel DS18B20 dg Adafruit IO
* Board   : NodeMCU ESP8266 V3
* Input   : 3 Ch DS18B20
* Output  : Adafruit IO
* 99 Proyek IoT
* www.ardutech.com

*****/

#include <ESP8266WiFi.h>
#include "Adafruit_MQTT.h"
#include "Adafruit_MQTT_Client.h"
#include <OneWire.h>
#include <DallasTemperature.h>

//---GANTI SESUAI DENGAN JARINGAN WIFI
//---HOTSPOT ANDA

#define WLAN_SSID       "ArdutechWiFi" // Nama Hotspot/WiFi
#define WLAN_PASS       "12345678"    // Password
#define AIO_SERVER       "io.adafruit.com"
#define AIO_SERVERPORT  1883           // use 8883 for SSL
//---GANTI SESUAI DENGAN USER NAME ADAFRUIT ANDA
#define AIO_USERNAME     "ArdutechJogja"
//---GANTI SESUAI DENGAN IO KEY ADAFRUIT ANDA
#define AIO_KEY           "aio_mAXJ90DJnEIBVj0IDmG6BIMZqMAW"
#define ONE_WIRE_BUS 14

WiFiClient client;

Adafruit_MQTT_Client mqtt(&client, AIO_SERVER, AIO_SERVERPORT,
AIO_USERNAME, AIO_KEY);

Adafruit_MQTT_Publish DS1 = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME
"/feeds/temp1");

```

```

Adafruit_MQTT_Publish DS2 = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME
"/feeds/temp2");

Adafruit_MQTT_Publish DS3 = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME
"/feeds/temp3");

OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);
int numberOfDevices;
float temperature1,temperature2,temperature3;
DeviceAddress tempDeviceAddress;
uint32_t x=0;
void MQTT_connect();
//=====
void printAddress(DeviceAddress deviceAddress) {
    for (uint8_t i = 0; i < 8; i++){
        if (deviceAddress[i] < 16) Serial.print("0");
        Serial.print(deviceAddress[i], HEX);
    }
}
//=====
void setup() {
    Serial.begin(9600);
    delay(10);
    sensors.begin();
    numberOfDevices = sensors.getDeviceCount();
    Serial.print("Locating devices...");
    Serial.print("Found ");
    Serial.print(numberOfDevices, DEC);
    Serial.println(" devices.");
    for(int i=0;i<numberOfDevices; i++){
        if(sensors.getAddress(tempDeviceAddress, i)){
            Serial.print("Found device ");

```

```

        Serial.print(i, DEC);
        Serial.print(" with address: ");
        printAddress(tempDeviceAddress);
        Serial.println();
    } else {
        Serial.print("Found ghost device at ");
        Serial.print(i, DEC);
        Serial.print(" but could not detect address. Check power and
cabling");
    }
}

Serial.println(); Serial.println();
Serial.print("Connecting to ");
Serial.println(WLAN_SSID);
WiFi.begin(WLAN_SSID, WLAN_PASS);
while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
}
Serial.println();

Serial.println("WiFi connected");
}
//=====
void loop() {
    MQTT_connect();
    sensors.requestTemperatures(); // Send the command to get temperatures
    for(int i=0;i<numberOfDevices; i++){
        if(sensors.getAddress(tempDeviceAddress, i)){
            float tempC = sensors.getTempC(tempDeviceAddress);
            if(i==0){
                temperature1 =tempC;

```

```

    Serial.print("Temperature 1: ");
    Serial.println(tempC);
  }
  else if(i==1){
    temperature2 =tempC;
    Serial.print("Temperature 2: ");
    Serial.println(tempC);
  }
  else if(i==2){
    temperature3 =tempC;
    Serial.print("Temperature 3: ");
    Serial.println(tempC);
  }
}

if (! DS1.publish(temperature1)) {
  Serial.println(F("Publish Channel 1 Failed"));
} else {
  Serial.println(F("Publish Channel 1 OK!"));
}

if (! DS2.publish(temperature2)) {
  Serial.println(F("Publish Channel 2 Failed"));
} else {
  Serial.println(F("Publish Channel 2 OK!"));
}

if (! DS3.publish(temperature3)) {
  Serial.println(F("Publish Channel 3 Failed"));
} else {
  Serial.println(F("Publish Channel 3 OK!"));
}

```

```
}  
    delay(5000);  
}  
//=====  
void MQTT_connect() {  
    int8_t ret;  
    if (mqtt.connected()) {  
        return;  
    }  
    Serial.print("Connecting to MQTT... ");  
    uint8_t retries = 3;  
    while ((ret = mqtt.connect()) != 0) { // connect will return 0 for  
connected  
        Serial.println(mqtt.connectErrorString(ret));  
        Serial.println("Retrying MQTT connection in 5 seconds...");  
        mqtt.disconnect();  
        delay(5000); // wait 5 seconds  
        retries--;  
        if (retries == 0) {  
            while (1);  
        }  
    }  
    Serial.println("MQTT Connected!");  
}
```

PERHATIKAN !

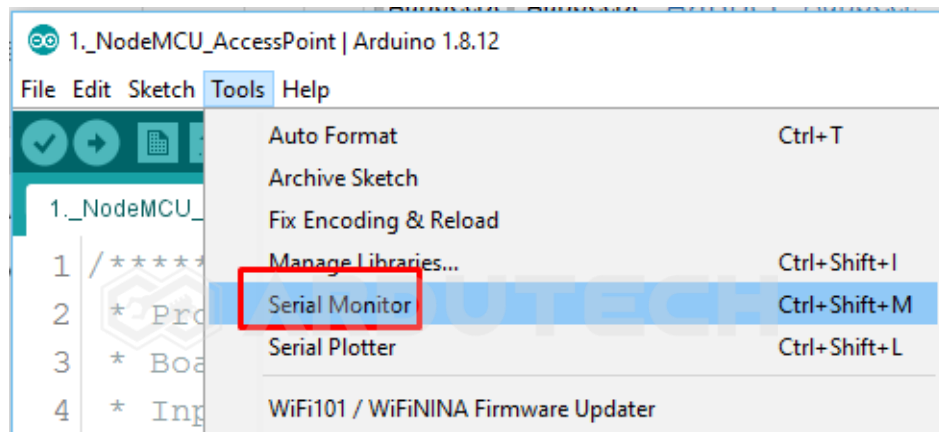
Ganti/sesuaikan variabel berikut :

- Nama jaringan WiFi/hotspot : **WLAN_SSID**
- Password jaringan WiFi/hotspot : **WLAN_PASS**
- Username Adafruit IO : **AIO_USERNAME**
- Active Key Adafruit IO : **AIO_KEY**

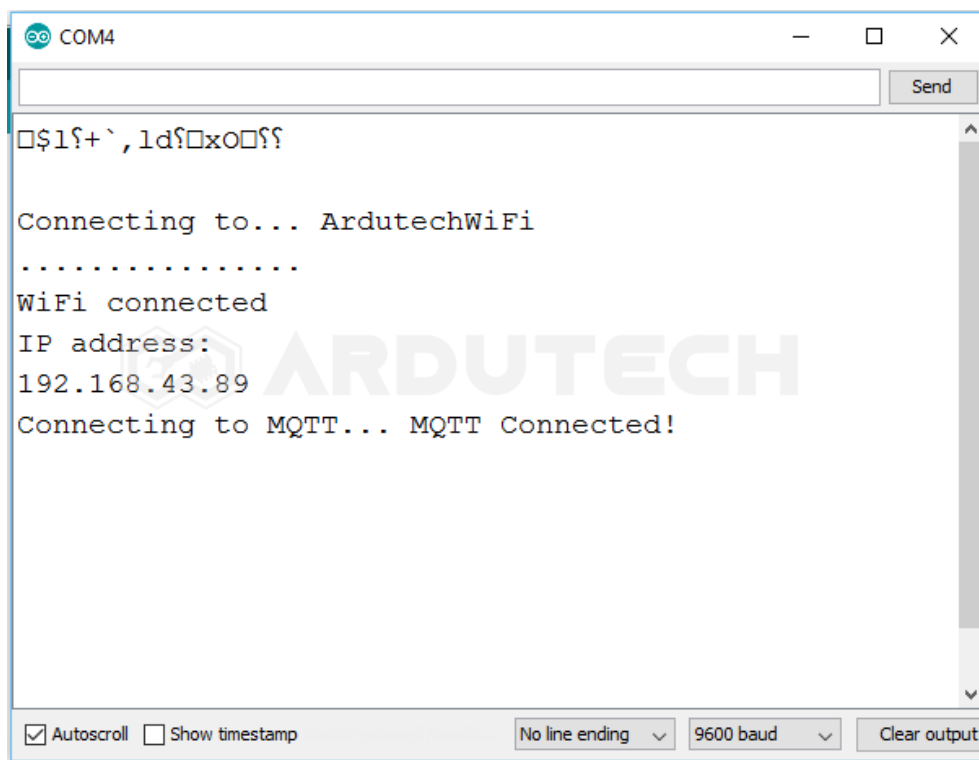
Simpan (**Save**) kemudian **Upload**. Pastikan tidak ada error, jika masih ada silakan cek penulisan dll kemudian perbaiki. (Program ini sudah diuji langsung dan sudah berjalan tanpa ada error)

Jalannya Alat

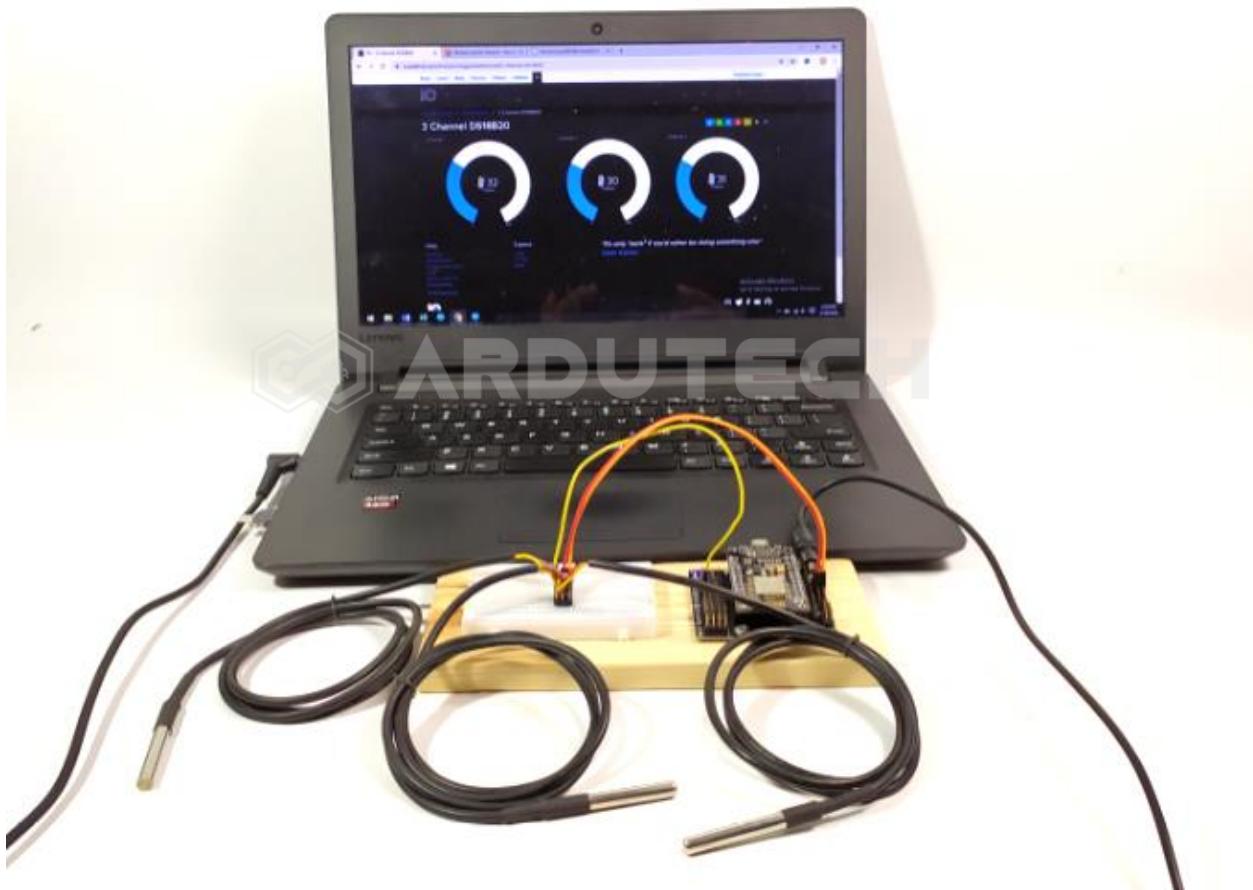
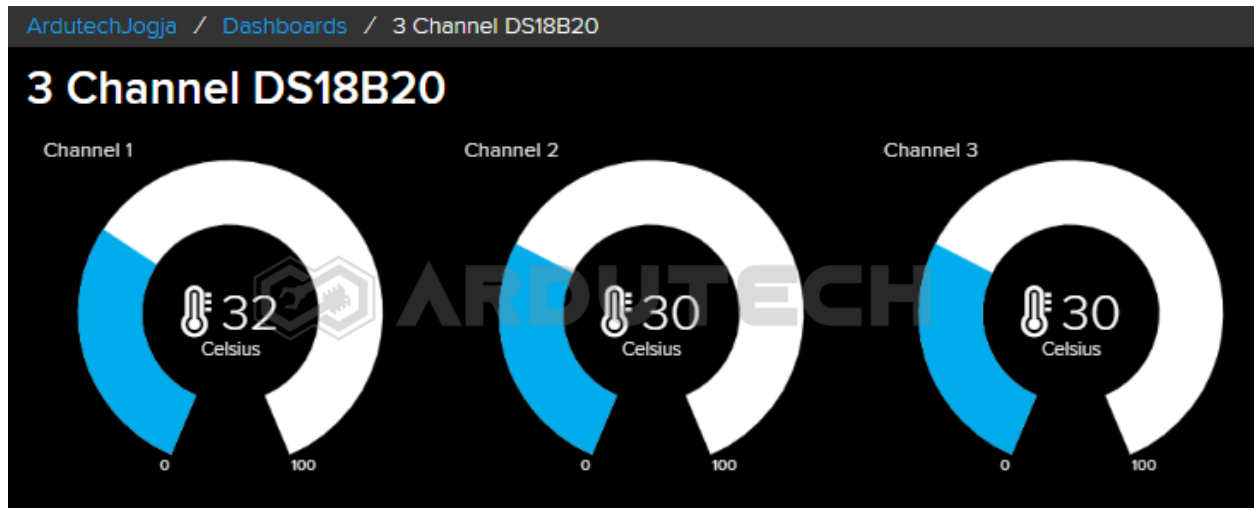
Siapkan jaringan WiFi/hotspot atau tethering dari HP dan aktifkan. Buka Serial Monitor, dari menu **Tools** → **Serial Monitor** kemudian seting baudrate pada nilai 9600.



Jika belum muncul tekan tombol reset (RST) pada board NodeMCU sehingga muncul tampilan sebagai berikut :



Setelah berhasil terhubung dengan jaringan WiFi sekarang kita kembali ke tampilan di Adafruit. Perhatikan nilai pada Gauge menunjukkan hasil pembacaan ke-3 sensor temperature DS18B20.



Beri perubahan suhu pada sensor DS18B20 kemudian perhatikan perubahan pada nilai Gauge-nya.

Selamat berkarya , semoga lancar dan bermanfaat.

Ardutech – *“Sahabat Inovasi Anda”*